

An Explicit Formula for the Loday Assembly

Virgil Chan

* Today Assembly

$$\alpha_L : BG_+ \wedge \mathbb{K}_R \longrightarrow \mathbb{K}_R[G]$$

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R = unital associative ring, G = discrete group

$\mathbb{K}_R$ = non-connective algebraic K-theory spectrum

BG = classifying space

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Homotopy groups

$$\text{RHS: } K_i(R[G]) := \pi_i(\mathbb{K}_{R[G]})$$

Algebraic K-theory of $R[G]$

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Algebraic K-theory of $R[G]$

{Hard to compute!}

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$$\alpha_L : BG_+ \wedge KR \longrightarrow KR[G]$$

Homotopy groups

LHS : $\pi_i(BG_+ \wedge KR)$ generalised homology theory

Atiyah-Hirzebruch Spectral Sequence :

$$E_{p,q}^2 \cong H_p(BG; K_q(R)) \Rightarrow \pi_{p+q}(BG_+ \wedge KR)$$

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$$\alpha_L : BG_+ \wedge KR \rightarrow KR[G]$$

Homotopy groups

More computable

LHS : $\pi_i(BG_+ \wedge KR)$ generalised homology theory

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Conjecture (K-theoretic Novikov Conjecture, Hsiang 1984)

If R is regular and G is torsion-free,
the map

$$\pi_i(\alpha_L) \otimes \text{id}_{\mathbb{Q}}: \pi_i(BG_+ \wedge K_R) \otimes \mathbb{Q} \longrightarrow K_i(R[G]) \otimes \mathbb{Q}$$

is injective for all $i \in \mathbb{Z}$. ▮

Thm (Bökstedt - Hsiang - Madsen 1993)

True if G is homologically finite. ▮

Conjecture (Classical Farrell-Jones Conjecture)

If R is regular and G is torsion-free, then

$$\pi_i(\alpha_L) : \pi_i(BG_+ \wedge K_R) \rightarrow K_i(R[G])$$

is an isomorphism for all $i \in \mathbb{Z}$.



Thm (Bartels-Lück-Reich)

True if G is word-hyperbolic.



There is a **subgroup** of $\pi_{1+g}(BG_+ \wedge KR)$

$$E_{1,g}^2 \cong G_{ab} \otimes K_g(R)$$

$$\downarrow$$

$$E_{1,g}^\infty$$



$$K_{1+g}(R[G])$$

$$\uparrow \pi_{1+g}(\alpha_L)$$

$$\pi_{1+g}(BG_+ \wedge KR)$$

which is a quotient of $E_{1,g}^2 \cong G_{ab} \otimes K_g(R)$.

Thm (C)

$$\begin{array}{ccc}
 E_{1,g}^2 \cong G_{ab} \otimes K_g(R) & \xrightarrow{\quad\quad\quad} & K_{1+g}(R[G]) \\
 \downarrow & \searrow \uparrow & \uparrow \pi_{1+g}(\alpha_L) \\
 E_{1,g}^\infty & \xrightarrow{\quad\quad\quad} & \pi_{1+g}(BG_+ \wedge K_R)
 \end{array}$$

The filler is given by

$$g \otimes [f] \mapsto [f] \star \{\mu_g\}.$$



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$\star$ = Loday product, $\{\mu_g\} \in K_1(\mathbb{Z}[G])$ multiplication by g^{-1} .

Corollary (Lower K-theory Cases)
 For a regular ring R , have:

$$G_{ab} \otimes K_0(R) \hookrightarrow \pi_1(BG_+ \wedge K_R) \xrightarrow{\pi_1(\alpha_L)} K_1(R[G])$$

$$g \otimes [P] \longmapsto \left\{ \eta_g: P \otimes R[G] \rightarrow P \otimes R[G] \right\}$$

$$u \otimes x \mapsto u \otimes x g^{-1}$$

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(Waldhausen, Lück-Reich)

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$$G_{ab} \otimes K_1(R) \hookrightarrow \pi_2(BG_+ \wedge K_R) \xrightarrow{\pi_2(\alpha_L)} K_2(R[G])$$

$$g \otimes \{u\} \longmapsto \{u, g^{-1}\}_{st} \quad \square$$

Corollary (The Case of $\mathbb{Z}[C_2^{\oplus n}]$)

The map

$$\pi_2(\alpha_L) : \pi_2(BC_2^{\oplus n} \wedge K_{\mathbb{Z}}) \longrightarrow K_2(\mathbb{Z}[C_2^{\oplus n}])$$

is

$$\begin{cases} \text{bijective} & \text{if } n=1 \text{ (Dunwoody),} \\ \text{injective} & \text{if } n>1. \end{cases} \quad \square$$

Note: Not surjective if $n=2$ (Yang-Tang-Liu).